Before we start discussing conditional probability etc. Let's just recall what have we learned about probability.

The discussion of probability starts with a random experiment which is an experiment whose outcome isn't known in advance. For our purposes, we agreed to consider experiments with finite number of possible outcomes only.

The set of all possible outcomes is called sample space and is usually denoted by S.

Any subset of S is called an event.

With all these terms, we defined probability using following three axioms.

i. For any event $E \subseteq S$, where S represents the sample space of a random experiment,

$$0 \leq Pr(E) \leq 1$$

ii. $Pr(S) = 1$

iii. If events $E_1, E_2, \cdots, E_n$ are pairwise disjoint or mutually exclusive, i.e., $E_i \cap E_j = \phi$, when $i \neq j$, then

$$Pr\left(\bigcup_{i=1}^{n} E_i\right) = Pr(E_1) + Pr(E_2) + \cdots + Pr(E_n)$$

Then finally we saw that if we have a random experiment where every experiment is equally likely, then for any event E, we have,

$$Pr(E) = \frac{|E|}{|S|}.$$

This is the summary of what we learned about probability so far. So let's start our discussion on conditional probability.

Conditional Probability:

Consider following random experiment.

We have a fair coin and we toss it twice.

Then the sample space, $S = \{HH, HT, TH, TT\}$

Now consider following events.

E := 'Both outcomes are Heads'

So, $E = \{HH\}$.

F := 'First outcome is Heads'.

So, $F = \{HH, HT\}$

and, G := 'At least one outcome is Heads'.

$G = \{HH, HT, TH\}$

As should be pretty obvious, we have

$$\Pr(E) = 1/4 \qquad \{ \ |E|/|S|$$

$$\Pr(F) = 1/2 \qquad \{ \ |F|/|S|$$

$$\Pr(G) = 3/4 \qquad \{ \ |G|/|S|$$

But now suppose I tell you that the result of our random experiment is actually event F, i.e., the first outcome is Heads, what will be the probability of event E now given this extra piece of information?

Well once I tell you that the event F has occurred, then practically the sample space actually shrinks down to F only.

And so now if event E <u>also</u> has to occur then we in fact have that the event $E \cap F$ will occur.

Therefore, probability of event E given that event F has occurred, which we denote as, $Pr(E|F)$, will be,

$$Pr(E|F) = \frac{|E \cap F|}{|F|}$$

which is same as,

$$Pr(E|F) = \frac{Pr(E \cap F)}{Pr(F)}$$

where, $Pr(F) \neq 0$.

Therefore, now you can verify that,

$$Pr(E|F) = 1/2$$

and $$Pr(E|G) = 1/3$$

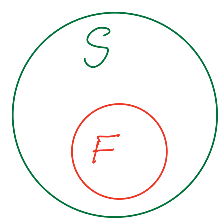
Some simple propositions related to conditional probability:

i. $Pr(S|F) = 1$, where $Pr(F) \neq 0$

Proof:

$$Pr(S|F) = \frac{Pr(S \cap F)}{Pr(F)}$$

$$= \frac{Pr(F)}{Pr(F)}$$



$$= 1$$

ii. For any two disjoint events A & B.

$$Pr((A \cup B)|F) = Pr(A|F) + Pr(B|F)$$

Proof: $$Pr((A \cup B)|F) = \frac{Pr((A \cup B) \cap F)}{Pr(F)}$$

$$= \frac{Pr((A \cap F) \cup (B \cap F))}{Pr(B)}$$

Since A & B are disjoint, do you see why $A \cap F$ & $B \cap F$ will also be disjoint?

You can convince yourself with examples etc, but another way to see is suppose $A \cap F$ & $B \cap F$ are not disjoint, then in fact A and B will also not be disjoint. Because the element that belongs to $A \cap F$ & $B \cap F$ will also belong to A & B.

So, Since $(A \cap F)$ & $(B \cap F)$ are disjoint,

Thus,

$$Pr((A \cup B)|F) = \frac{Pr(A \cap F)}{Pr(F)} + \frac{Pr(B \cap F)}{Pr(F)}$$

$$= Pr(A|F) + Pr(B|F)$$

where the second last equation above follows due to axiom 3 of probability.

iii. $Pr(E^c|F) = 1 - Pr(E|F)$

Proof: We know,

$$S = E \cup E^c$$

So, $1 = Pr(S|F) = Pr(E \cup E^c|F)$

$$= Pr(E|F) + Pr(E^c|F)$$

$$\therefore Pr(E^c|F) = 1 - Pr(E|F).$$

Multiplication rule of probability.

We learned that,

$$Pr(E|F) = \frac{Pr(E \cap F)}{Pr(F)}$$

$$\Rightarrow \quad Pr(E \cap F) = Pr(F) \cdot Pr(E|F)$$

Similarly, $Pr(F \cap E) = Pr(E) \cdot Pr(F|E)$

This is what we call Multiplication Rule of Probability.

In general, we have;

for $E_1, E_2, \cdots, E_n$, a sequence of n events,

$$\Pr\left(\bigcap_{i=1}^{n} E_i\right) = \Pr(E_1)\cdot\Pr(E_2|E_1)\cdot\cdots\cdot\Pr(E_n|E_1\cap\cdots\cap E_{n-1})$$

Independent Events:

As said by someone in the class,

Two events are said to be independent, if occurrence of one doesn't affect other.

formally, for any two events E & F, we say that they both are independent, if

$$\Pr(E) = \Pr(E|F)$$

and,

$$\Pr(F) = \Pr(F|E)$$

Since, $\Pr(E|F) = \dfrac{\Pr(E\cap F)}{\Pr(F)}$

$$\Rightarrow \quad \Pr(E) = \frac{\Pr(E\cap F)}{\Pr(F)}$$

$\{ \Pr(E|F) = \Pr(E)$

So, $$\Pr(E\cap F) = \Pr(E)\cdot \Pr(F)$$

* Suppose events E and F are independent. Then will the events E^c & F also be independent?

The answer is yes.

Proof: To show E^c & F will be independent given that E & F are independent, all we need to show is that,

$$\Pr(E^c\cap F) = \Pr(E^c)\cdot \Pr(F)$$

By multiplication rule,

$$\Pr(E^c\cap F) = \Pr(F\cap E^c)$$

$$= \Pr(F)\cdot \Pr(E^c|F)$$

$$= \Pr(F)\cdot(1-\Pr(E|F))$$

$$= \Pr(F)\cdot(1-\Pr(E))$$ $\left\{\text{Since } E \text{ \& } F \text{ are ind.}\right.$

$$= \Pr(F)\cdot \Pr(E^c)$$ $\blacksquare$

Similarly, we can show that,

Given E & F are independent,

- E & F^c will be independent.
- E^c & F^c will be independent.